Gears and wheels

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use gear ratio and wheel circumference to predict an ideal travel speed.

SUCCESS CRITERIA - I can...

1. Connect gear teeth with speed ratio.
2. Calculate distance per wheel revolution.
3. Explain the tradeoff between ideal speed and available torque.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gear ratio” and “circumference” to explain your answer. Sentence starters are optional.

gear ratio: a comparison of gear tooth counts or rotation rates

circumference: distance around the edge of a circle

torque: the turning effect of a force about an axis

01 Warm-up: bring back an earlier idea

A motor uses **3 W** for **5 s**. **Find electrical energy** and **name a possible non-mechanical output**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

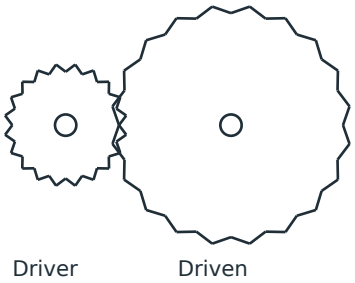
02 Worked example: explain the prediction

Reference: $rpm_{out} = rpm_{in} \times teeth_{driver}/teeth_{driven}$; $C = \pi d$
$v = C \times rpm_{wheel}/\mathbf{60}$.
Ideal gears and rolling without slip; C in meters gives v in m/s. Loading and losses change real performance.

Worked reasoning: A **10**-tooth driver turns a **30**-tooth driven gear. The driven gear makes one turn for every three driver turns. In an ideal gear pair, slower rotation trades for greater torque.

Your turn: Draw both gears, label tooth counts and opposite rotation directions.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

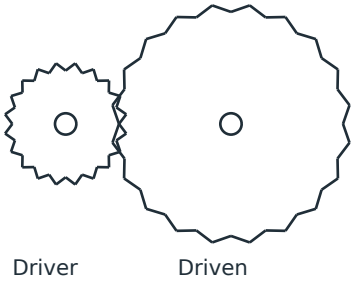


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A small driver gear turns a larger driven gear. **Predict which turns more slowly** and indicate their opposite rotation directions.

Sentence starter: Both ... . They differ in ... because ... .



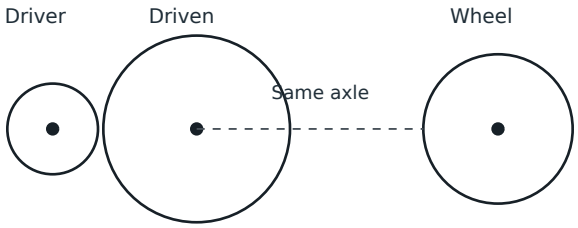
Marzano check - circle: ① ② ③

04 Apply the physics

A small classroom car’s wheel circumference is **0.20 m** and its wheel rate is **120 rpm**. **Label the wheel diagram**. Convert rpm to revolutions per second, then **calculate ideal ground speed** with no slip.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found ... [unit]. This value means ... in this situation.



Label teeth, rotation rates and wheel size.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A team chooses the largest wheels because the formula predicts more distance per turn. **Explain why measured acceleration or** speed might still decrease.

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A total current of **14 A** splits between two identical parallel branches. **Find each branch current**.
Draw the stated circuit and **label quantities relevant** to the question.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw the complete circuit and label shared nodes.

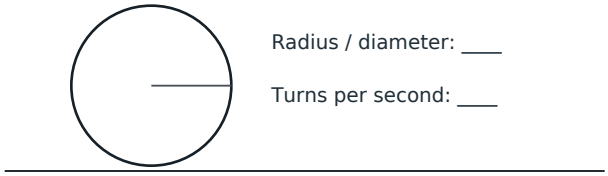
Marzano check - circle: 1 2 3

07 Exit check

At the same wheel rpm, doubling circumference does what to ideal travel speed? **State one real-world limitation**.

Word help: ideal = using stated simplifying conditions

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



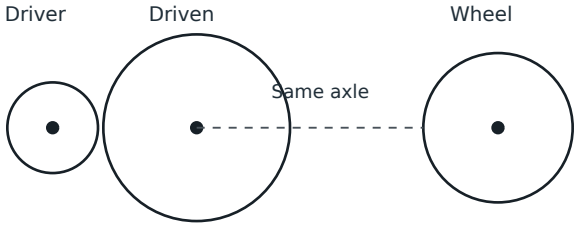
Travel per turn: ____ Ground speed: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Design two different tooth-count pairs giving the same **3**: **1** reduction.
2. **Explain whether** tooth counts alone find the car's ground speed.

Sentence starter: Both ... . They differ in ... because ... .



Label teeth, rotation rates and wheel size.

Marzano check - circle: 1 2 3